

AARYAN SHANBHAG

aaryan6104@gmail.com • +65 9144 4030 • [LinkedIn](#) • [GitHub](#)

SUMMARY

Engineering Systems Design student at SUTD specialising in Business Analytics, with hands-on experience in time-series forecasting, NLP pipelines, and decision-support modelling across industry and international field projects. Proficient in Python, SQL, Excel, and Tableau.

EDUCATION

Singapore University of Technology and Design

B. Eng. in Engineering Systems Design

Sept 2024 – Present

Tentative Specialisation: Business Analytics and Operations Research

Global Indian Int'l School, Punggol

International Baccalaureate (IB) Diploma

June 2020 – May 2022

COURSEWORK

40.011 – Data & Business Analytics (SBSTransit Industry Project)

- **Built predictive maintenance pipeline** for SBSTransit railway track degradation using **segmented regression** (Python Statsmodels & Matplotlib).
- Automated **multi-sheet preprocessing** and **feature engineering** using **Excel Macros & VBA**.
- Designed **decision-support system** predicting time-to-failure and maintenance urgency.

GEO ITS x SUTD – Overseas Field Study (Surabaya, Indonesia)

- **Forecasted fuel consumption and revenue** for Surabaya's public bus operator using **Holt-Winters**.
- Built **3-scenario forecast** (optimistic/moderate/pessimistic) with 1.96 std. deviation confidence intervals; **Benefit-Cost Analysis (BCA = 0.93)** indicates if an EV fleet transition is viable in the 5-year horizon.
- Modelled time-series data in **Python/Jupyter** and **Excel**; evaluated via **MAPE & RMSE**.
- Conducted **field observations** to link operational inefficiencies with data patterns.

02.137DH – Digital Humanities

- Built **end-to-end NLP pipeline in Python/Jupyter** using **spaCy dependency parsing** to extract 191 labelled verb-instances from unstructured news text into structured **pandas DataFrames**.
- **Engineered custom rule-based extractors** for grammatical voice, epistemic hedging, actor classification and lexical framing, operationalizing Critical Discourse Analysis theory as **computable metrics**.
- Applied **chi-square significance testing (scipy.stats)** and produced visualisations in Matplotlib, Seaborn to compare and quantify linguistic framing asymmetries across news outlets.

40.002 – Optimisation

- Developed **MILP-based ER patient triage model** for a hospital, optimising admission decisions (Discharge/ICU/Ward) with capacity and routing constraints using **Julia | JuMP | HiGHS**.
- Implemented **Optimal Classification Trees** with branch/leaf/big-M constraints; performed scenario analysis under demand surges and capacity limits to evaluate model robustness.

TECHNOLOGIES

Excel – Pivot Tables, Macros, VBA, Analysis ToolPak

SQL – Table Join, Excel-Python integration

Python – pandas, Matplotlib, statsmodels, sk-learn
SciPy, spaCy, Seaborn

R – tidyverse (dplyr, ggplot2)

Julia – JuMP, HiGHS

Tableau, PowerBI – DAX

THEORY

Financial Analysis and Forecasting:

Cash Flows | linear/logistic regression | NPV/IRR metrics
Holt-Winters (exp. smoothing) | DuPont decomposition

Statistical Inference and Hypothesis Testing:

Bayesian inference | Confidence Intervals | chi 2 & t-tests
ANOVA | bootstrap resampling | non-parametric tests

Optimisation – LP/MILP | Simplex | Branch-and-Bound
Dynamic Programming | Primal-Dual & Lagrangian duality